H. ASHEN KAVISHA

Information Technology Undergraduate (specialized in Data Science)

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Ashen Kavisha

Ashen Kavisha Lakshan

Profile Summary

Second-year IT undergraduate at SLIIT specializing in Data Science, with hands-on experience in full-stack web development, mobile applications, and machine learning. Built and deployed production projects including a live freelance platform and an ML classification system for lung cancer prediction. Proficient in Python, React Native, Node.js, and MongoDB, with a strong foundation in data preprocessing, feature engineering, and model training. Passionate about building scalable, data-driven applications that solve real-world problems.

Main Projects (Click the titles to access the projects)

Smart Vehicle Parking Management System

|Jan 2025 – Jun 2025|

Tech Stack: HTML, CSS, JavaScript, MsSQL

- Developed a comprehensive web application for managing vehicle parking based on vehicle type with real-time availability tracking.
- Utilized data structures and algorithms to optimize parking slot allocation and vehicle availability checking.
- Designed responsive and intuitive user interface using HTML5, CSS3, JavaScript, and Bootstrap framework Built secure user authentication and donor eligibility validation workflows.
- Implemented backend functionality using Node.js/Express.js with restful API architecture.
- Designed and managed MS SQL database schema with normalized tables for efficient data storage and retrieval

Hospital Management System

|July 2025 – November 2025|

Tech Stack: java, javaScript,MsSQL , HTML,CSS

- Simplified the interaction between patients and doctors through a clean, role based interface.
- Provided a fast, user-friendly booking system interface for patients to search and book appointments.
- Ensured data security with role-based authentication and data integrity.
- Enabled doctors to manage their availability and view appointment schedules assigned.

- Designed a clean UI for seamless navigation and improved user experience.
- Allowed online booking and emergency appointment requests .

Lung Cancer Prediction System **Machine Learning**

|July 2025 – November 2025|

Tech Stack: Python, Pandas, NumPy, Scikit Learn, Seaborn

- Developed ML classification system with optimized feature engineering pipeline using SelectKBest and mutual information scoring, achieving dimensionality reduction from full feature set to top 10 predictive features.
- Implemented end-to-end data preprocessing workflow including class balancing, normalization, and correlation analysis (threshold >0.85) to ensure high-quality training data for medical prediction tasks.
- Built reproducible Python pipeline using Scikit-learn, Pandas, and Seaborn for automated feature selection and statistical analysis of patient health data.

Skill Lab – Online Freelance Platform (MERN)

Tech Stack: MongoDB, Express.js, React.js, Node.js

- Built a full-stack freelance marketplace platform supporting job posting, bidding, and project management. □ Developed user registration, service listing, secure messaging, and profile management modules.
- Implemented backend logic and API routes using Express and Node.js with MongoDB as the data layer.
- Designed a responsive UI in React enabling smooth real-time interactions between freelancers and clients.

Assignment service web application

|December 2025 – February 2026|

Tech Stack: MsSql, javaScript, Html,Bootstrap,Maven

- Created an assignment service platform for submit assignments based on user needs.
- Users can submit assignments , and get solution delivery via email and users dashboard notification inbox.
- Created an admin panel to receives the assignments that submit by users. And it receives via email to the administrator .
- Administrator approve the assignment and send the solutions to the user via email, user heads to the payment gateway and pay for the service.
- Used Mssql database schema with enhanced tables with users credentials security.
- Database Hosted on Microsoft Azure services.
- Web site is now live take a look at - https://www.assignmentservice.net

E-Library Mobile App

|January 2026 – June 2026|

Tech Stack: React Native,Expo SDK,NODE + EXPRESS.js ,MongoDB,JSON

- This application is a cross platform mobile application that allows read books to the users.
- Created an favourite bookshelf to access the saved books.
- Users can manage a personal bookshelf with reading lists(Reading now, Favourites , Wishlist)

- Favourites individual books for quick access
- Also this contains AI powered book recommendations .
- USED Min-Max normalisation — puts rating and year on the same 0–1 scale as the similarity score so they can be added together fairly
- Also added Weighted score fusion — the core formula that combines all three signals (similarity, rating, recency) with tunable weights. And Metadata filtering — hard rules that remove books before ranking even begins (rating, year, author, language)

IPO Listing Profitable Prediction

|FreeTime(Own) Project Jun-2026 |

Tech Stack: Python – TensorFlow / Keras , Pandas , Numpy, Sckit-Learn

- This Project Contains A Indian IPO would list profitably, using a dataset of ~320 IPOs. (Small Dataset).
- Removed Outliers Using IQR Clipping and Min-Max Scaling to balance the data that most important to change behaviour of Target Feature.
- Also added correlation heatmap so can identify which feature affects target feature. And less affect features removed.
- Splitted the dataset and achieved 72% test accuracy; also got the idea that accuracy can be increase via adding more data.
- Gained hands-on experience in the deep learning workflow: architecture design, regularization trade-offs, and the relationship between data quality/quantity and model generalization.

Car Sales Data Analysis (Real World Project)

|FreeTime(Own) Project Jun-2026 |

Tech Stack: Python,Pandas ,jupyter Notebook

- Performed exploratory data analysis on a large used-car listings dataset(Over 35000+ records) using Python and Pandas.
- Cleaned and transformed raw data by handling outliers and removing irrelevant attributes.
- Analyzed vehicle pricing and mileage trends across major automobile brands.
- Calculated brand-wise average prices and mileage statistics to identify market patterns.
- Created visualizations to communicate analytical findings.

Image Classification – Dog Breed Identifier(CNN)

|FreeTime(Tested Project Jun-2026 |

Tech Stack: Python,Pandas ,jupyter Noteboook,keras/tensorflow,PIL

- Built a multi-class CNN image classifier to identify dog breeds from a dataset of 926 labled images using tensorflow/Keras.
- Iteratively improved model architecture from a baseline single-layer CNN to a 3-block convolutional network with max pooling and dropout regularization to reduce overfitting.
- Implemented data augmentation (random flip, rotation, zoom) to improve generalization on a limited-size dataset.
- Built an evaluation pipeline comparing predicted vs. actual labels with visual inspection of misclassified samples to diagnose model weaknesses.

CI/CD DevOps Pipeline

|June 2026 – July 2026|

Tech Stack: React/Vite, Node.js / Express, PostgreSQL, Docker, Jenkins, SonarQube, AWS EC2, Nginx

- Designed and deployed a fully automated CI/CD pipeline across three dedicated AWS EC2 instances, triggered automatically via GitHub webhooks on every push.
- Implemented a Jenkins declarative pipeline covering dependency installation, frontend build, static code analysis (SonarQube), Docker image builds, and automated remote deployment via SSH.
- Containerized a full-stack application (React frontend, Express backend, PostgreSQL database) using multi-stage Docker builds, reducing final frontend image size by serving static assets through Nginx rather than a full Node.js runtime.
- Configured Docker Hub as the image registry and built an automated build-push-pull-deploy workflow, eliminating manual deployment steps entirely.
- Used Nginx and Jenkins infrastructure constrains to execure scheduling and dis monitoring thresholds.
- Diagram: GitHub → Jenkins (build & test) → SonarQube (code quality gate) → Docker Hub → AWS EC2 (live deployment).

Education

2024 - Present

BSc (Hons) in IT specializing in Data Science Sri Lanka Institute of Information Technology(SLIIT)

2023 G.C.E A/L in Physical Science stream

- Physics-C
- Combined Mathematics-C
- Chemistry-C

Certifications

- **The Full Stack Web Development Bootcamp 2023-MERN STACK**
- **Numpy For Data Science Real Time Experience**

Skills

Programming	*CLOUD & DevOps*	*Web*	*Database*	*Tools & Others*
• Python	• AWS	• Node.js	• MySQL	• Github
• Java	• Azure	• React	• SQL	• Git
• JavaScript	• Docker	• React Native/Vite	• MongoDB	• Postman
• C#	• Jenkins	• Express.js	• MsSql	• Github Actions
• Kotlin	• SonarQube	• Next.js	• PostgreSQL	• Github Webhooks
• C++	• Nginx	• HTML		• SpringBoot
		• CSS		• TomCat
		• Bootstrap		• Adobe Photoshop
		• WordPress		• CapCut
				• Adobe Premiere Pro

Languages

- English
- Sinhala

Reference

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